Grid event or data artefact? Hierarchical weak supervision for anomaly attribution in multi-city feeder telemetry

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Abstract

[Written last: at most 250 words; purpose, procedures, main findings with their statistical significance, principal conclusions.]

Keywords: weak supervision, anomaly attribution, feeder telemetry, data quality, outage statistics, load forecasting, distribution networks

1. Introduction

[To be written.]

2. Related work

[To be written.]

3. Data, hierarchy and typology

3.1. Datasets

The study uses the UCI high-resolution load dataset from smart meters in four Moroccan cities [1, 2], introduced with a load-forecasting study [3]. The meters are installed by the national utility ONEE on 22 kV distribution feeders and record at 10-min intervals (30 min in Marrakech); the data were exported manually from the meters [2]. In Laâyoune, Boujdour and Foum El Oued the dataset covers every 22 kV feeder departure [2], so each city is observed in full. Each feeder is a *zone* in the dataset's terminology.

We analyse the file `Data Morocco.xlsx` of the UCI record (22,434,854 bytes; SHA-256 in the data availability statement), which is byte-identical to

Table 1: Datasets. Regular steps: length of the regular time grid from the first to the last stamp. Missing steps: grid steps without a record. Zone-steps: one zone at one time step. Exact zeros are counted after duplicated stamps are averaged (Foum El Oued has 4,329 exact zeros in the raw rows).

City	Zones	Step (min)	Unit	Period	Regular steps	Missing steps	Exact-zero zone-steps
Laâyoune	5	10	A	Sep 2022 – May 2024	88,890	12	1,465
Boujdour	3	10	A	Sep 2022 – May 2024	88,890	12	1,413
Foum El Oued	7	10	A	Sep 2022 – May 2024	88,890	12	4,323
Marrakech	2	30	kW	Jan 2023 – Jan 2024	17,520	25	101
Tetouan[a]	3	10	n.s.	Jan – Dec 2017	52,416	0	0

[a]Transfer test only. n.s.: unit not stated in the repository record [4].

the file the repository served on 14 November 2026. Table 1 lists the series. The three southern cities (15 zones, in amperes) share one 10-min timeline from 14 September 2022 to 24 May 2024. In the data paper's table, Foum El Oued zones 5–7 end on 26 December 2023 [2]; in the analysed file they continue without a break until 24 May 2024. We call the three cities the *southern system*; it carries most of the analysis. Marrakech (2 zones, in kW, 2023) is recorded on its own 30-min timeline and is treated as a separate system. The UCI Tetouan dataset [4, 5], three distribution networks of one city at 10-min steps in 2017, is used only to test transfer to a new system (Section 6). Fig. 1(a) locates the five cities.

The repository record lists no missing values [1], but the regular time grid lacks the same 12 stamps in every southern zone and 25 in Marrakech, and some stamps occur twice: 12 per southern city, all on 30 October 2022 and 29 October 2023, and 3 in Marrakech. Duplicated stamps are averaged. None of the datasets comes with event labels: there is no record of which zeros or flat stretches are outages and which are acquisition failures. Exact zeros alone occupy 7,201 zone-steps in the southern system (Table 1). All time stamps are used as recorded, without any clock correction, so that clock errors remain observable as a data property.

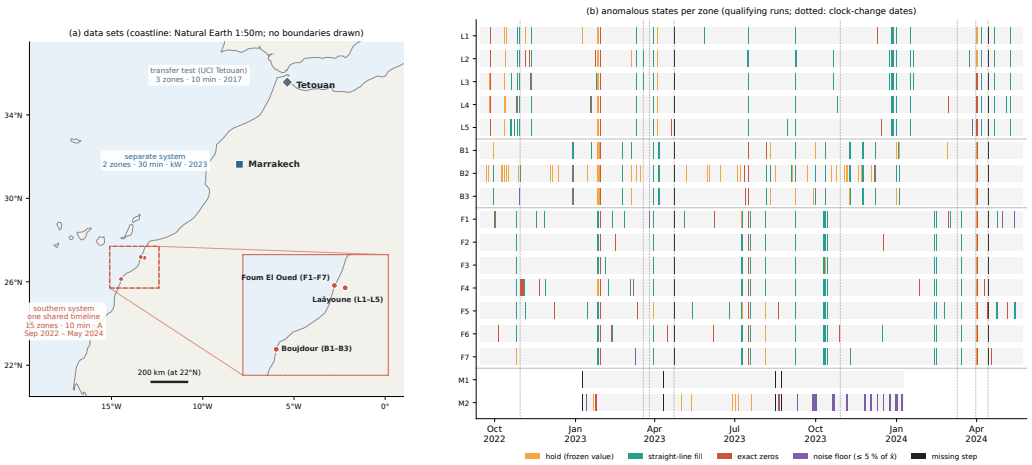


Figure 1: Data. (a) The five cities. Laâyoune, Boujdour and Foum El Oued (15 zones, 10 min, A) share one recording timeline; Marrakech (2 zones, 30 min, kW) is a separate system; Tetouan (3 zones, 2017) is the transfer test. Coastline from Natural Earth (1:50m, public domain); no boundaries are drawn. (b) Anomalous states per zone (holds, straight-line fills, exact zeros and noise-floor runs of qualifying length, and missing steps; Section 4); dotted lines mark clock-change dates. Time stamps as recorded.

3.2. The telemetry hierarchy

The southern series form a three-level hierarchy: zone (one feeder), city (all feeders of one city) and region (the three cities). Each level corresponds to a different cause of a disturbance. A feeder fault or planned maintenance affects one zone. Losing a city's supply point de-energises all of that city's zones while the other cities keep running. A regional event affects several cities at once.

The acquisition chain forms a fourth, non-electrical layer. The three cities are recorded on one timeline, and a failure in the shared recording or export process can hit zones in different cities at the same time step, regardless of the electrical topology. We hypothesise that such failures leave traces that a de-energised feeder does not: series held at an identical value, stretches that are perfectly straight lines, and exact zeros, where a de-energised feeder would read a small, noisy residual current. The method treats these signatures, together with the level at which an anomaly appears, as evidence: simultaneity across cities combined with these traces points to an artefact, whereas a noise floor confined to one zone or one city points to a grid event. Section 6 tests this hypothesis on the real data.

3.3. Problem formulation

Let $x_{z,t}$ be the recorded value of zone z at time step t, and P the number of steps per day (144 at 10 min, 48 at 30 min). The expected value is the median of the same time-of-day slot over the previous $K = 14$ days, excluding zeros:

$$\hat{x}_{z,t} = \text{median}\left\{x_{z,t-kP} \;:\; k = 1, \ldots, K,\; x_{z,t-kP} \neq 0\right\}. \tag{1}$$

During the first 14 days of each series, $\hat{x}$ rests on fewer than K days, and this period is treated as a warm-up. The relative level

$$\rho_{z,t} = x_{z,t} / \hat{x}_{z,t} \tag{2}$$

is unit-free, so amperes, kilowatts and feeders of different sizes are compared on one scale.

An *item* i is a candidate event: a dead or fill span with its set of affected zones, a clock-change date in a city, or a load-transfer day for a pair of zones (Section 4.2). Items that touch or follow one another are linked into *episodes*, such as a hold followed by a zero fill. The type of an item y_i belongs to $\mathcal{Y} = \{\text{A1}, \ldots, \text{A4}, \text{G1}, \ldots, \text{G5}, \text{N}\}$ (Section 3.4). The task has four parts: *detection*, finding the items; *attribution*, estimating a probability distribution $p_i(y)$ over $\mathcal{Y}$ for each item; *localisation*, identifying the level (zone, city or region) that carries the anomaly; and *type-aware cleaning*, imputing artefacts while keeping grid events in the data and marking them. All four must be solved without labelled examples.

3.4. Typology

Table 2 and Fig. 2 define nine anomaly types, each with a measurable signature at one level of the hierarchy. The four *data artefacts* are zero-fill gaps across cities (A1), simultaneous fills of several zones by held values or straight lines (A2), the same fills in a single zone (A3), and clock errors that shift the daily profile after a clock-change date (A4). The five *grid events* are regional outages (G1), city outages (G2), unplanned feeder outages (G3), planned work (G4) and load transfers between feeders (G5). Items that no type fits are labelled N (normal or untyped).

Straight-line fills were not part of the initial typology. They appeared in the data audit (Section 6) and are now part of A2 and A3. The expected signatures of grid events, in particular gradual restoration and cold-load pick-up after an outage, are hypotheses that Section 6 tests on the real data.

Table 2: Typology. Level: the level of the hierarchy at which the signature appears. Rules: the labelling rules that vote for the type (Section 4). Real items: rule items assigned the type in the final catalogue of the real data. η: share of exact zeros in a dead run; π: restoration overshoot; δ^*: estimated clock offset; κ: conservation index.

Type	Name	Level	Signature	Rules	Real items
A1	Zero-fill gap	Region (≥ 2 cities)	Exact zeros ($\eta = 1$) in ≥ 2 cities at once, often after a hold or straight-line fill; returns at once	R1, R3, R4, R21	3
A2	Simultaneous fill	≥ 2 zones	≥ 2 zones held at identical values or straight-line interpolated over the same span	R2, R19	57
A3	Single-zone fill	Zone	One zone held or straight-line interpolated while its neighbours move normally	R12, R20	98
A4	Clock error	System, one day	Daily profile shifted by $\delta^* \approx \pm 60$ min after a clock-change date; duplicated or missing stamps	R13, R14, R15	14
G1	Regional outage	Region (≥ 2 cities)	Noise floor ($\eta < 0.9$) in ≥ 2 cities; hypothesised: gradual restoration with cold-load overshoot $\pi \geq 1.15$	R5, R6, R7	0
G2	City outage	City	All zones of one city at the noise floor while the other cities stay normal	R8 (R5, R6)	1
G3	Unplanned feeder outage	Zone	One zone at the noise floor at any hour, possibly for more than 24 h	R9, R11 (R6)	52
G4	Planned work	Zone	One zone at the noise floor, starting 06:00–11:59, shorter than 6 h	R10	70
G5	Load transfer	≥ 2 zones of a city	One zone steps down while another steps up by the same amount ($\kappa \approx 0$)	R16	1

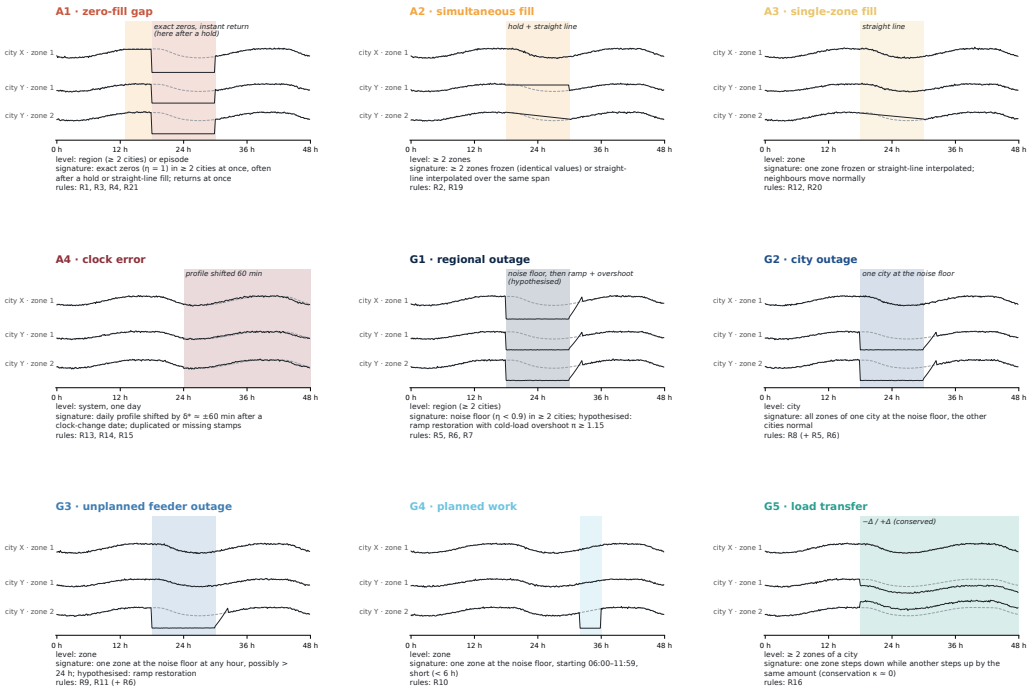


Figure 2: Typology. Four data artefacts (A1–A4) and five grid events (G1–G5), each with a measurable signature at one level of the zone–city–region hierarchy and the rules that vote for it (schematic). Straight-line fills are part of A2 and A3.

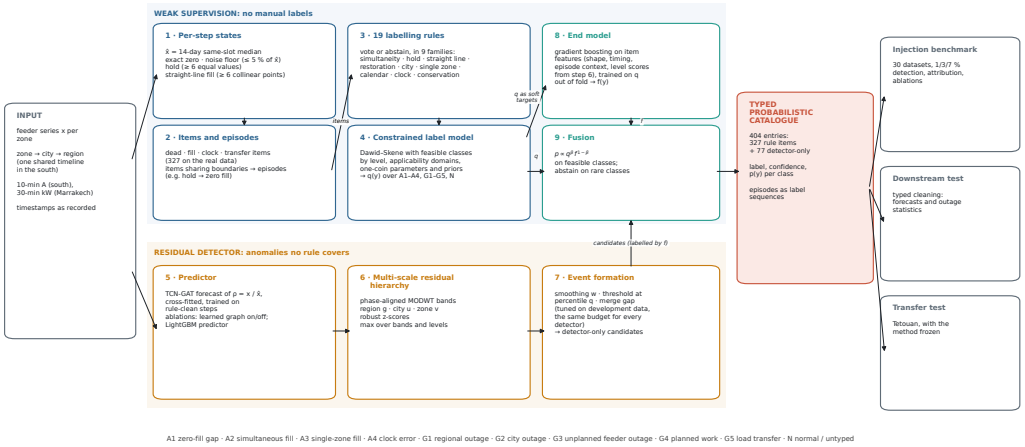


Figure 3: Framework. Per-step states and items feed 19 labelling rules and a constrained Dawid–Skene label model (weak supervision, no manual labels). A multi-scale hierarchical residual detector finds anomalies that no rule covers. An end model and a fusion step produce the typed, probabilistic catalogue, which is evaluated by injection, by a downstream test and by transfer to another city (Section 5).

4. Method

4.1. Overview

Fig. 3 summarises the framework. Per-step states turn each series into candidate *items*. Labelling rules vote on the items, and a label model combines the votes into probabilistic labels without any manual annotation (weak supervision [6]). In parallel, a multi-scale residual detector scores every zone at every step and finds anomalies that no rule covers. An end model learns the labels from item features, including the detector's scores, and a fusion step combines both sources of evidence into the final typed, probabilistic catalogue. Type-aware cleaning then uses the catalogue.

4.2. Per-step states, items and signatures

Four states are computed for each zone series. A step is an exact zero if $x_{z,t} = 0$ and at the noise floor if $x_{z,t} \le \varepsilon\, \hat{x}_{z,t}$, with $\varepsilon = 0.05$. A zone is *dead* at step t when

$$d_{z,t} = \mathbb{1}[x_{z,t} = 0] \;\vee\; \big(\mathbb{1}[x_{z,t} \le \varepsilon\, \hat{x}_{z,t}] \wedge \neg \ell_{z,t}\big), \tag{3}$$

where $\ell_{z,t}$ marks straight-line fills (below); dead runs need at least three steps. For a dead run E, the exact-zero share

$$\eta_E = \frac{1}{|E|} \sum_{t \in E} \mathbb{1}[x_{z,t} = 0] \tag{4}$$

separates exact zeros, which a recording gap produces, from the small residual current of a de-energised feeder. A step is *held* if it belongs to a run of at least m_h identical non-zero values,

$$h_{z,t} = \mathbb{1}[x_{z,t} = x_{z,t-1} \neq 0], \tag{5}$$

with $m_h = 6$ at 10 min and 12 at 30 min. Straight-line fills, the signature found in the data audit, are runs of at least m_ℓ collinear points (6 at 10 min, 12 at 30 min), detected through a vanishing second difference:

$$\ell_{z,t} = \mathbb{1}\big[\, |\Delta^2 x_{z,t}| < 10^{-8} \max(1, |x_{z,t}|) \;\wedge\; |\Delta x_{z,t}| > 10^{-12} \;\wedge\; x_{z,t} \neq 0 \,\big]. \tag{6}$$

When states overlap, exact zero takes precedence over a straight-line fill, which takes precedence over the noise floor and then the hold.

Items are the units the rules vote on. Dead runs of one system whose starts lie within $\tau = 10$ min of the first start form a dead item; hold and straight-line runs form fill items in the same way. The simultaneity of an item is the number of cities involved,

$$S_E = \left|\{c : \exists z \in c,\ |s_z - s_E| \leq \tau\}\right|, \qquad s_E = \min_z s_z, \tag{7}$$

where s_z is the start of zone z's run. Each item also records whether a fill run in another zone is co-extensive with it (time intersection over union ≥ 0.5) and whether a fill ends exactly where a dead run begins. Items that overlap or touch, and share a boundary or most of their span, are linked into *episodes*.

Restoration is measured after the event, but only once any straight-line fill that follows it has ended, because such a fill imitates a ramp. With k_z the first step after the run and after any adjoining fill in zone z,

$$r_E = \frac{\sum_z \sum_{j=0}^{n_{30}-1} x_{z,k_z+j}}{\sum_z \sum_{j=0}^{n_{30}-1} \hat{x}_{z,k_z+j}}, \qquad \pi_E = \max_{0 \leq j < n_{60}} \frac{\sum_z x_{z,k_z+j}}{\sum_z \hat{x}_{z,k_z+j}}, \tag{8}$$

where n_{30} and n_{60} are the steps in 30 and 60 min; the ramp time τ_E is the delay until the ratio first reaches 0.9 (within 6 h).

Two further item kinds test the remaining types. For load transfers, $\Delta_z(d)$ is the change in a zone's 14-day median of daily means around day d, and a candidate day is one where one zone rises and another falls by more than a quarter of their mean level. The conservation index

$$\kappa(d) = \frac{\left|\sum_z \Delta_z(d)\right|}{\max\left(\Delta^{+}(d),\, -\Delta^{-}(d)\right)}, \tag{9}$$

with Δ^{+} and Δ^{-} the largest rise and fall, is near 0 when current moves between feeders and near 1 when load appears or disappears. For clock errors, candidate dates are the European and Moroccan clock-change dates and every date with duplicated or missing stamps. With p_{pre} and p_{post} the mean normalised daily profiles over the 7 days before and after a date (anomalous steps removed), the clock offset is

$$\delta^{*} = \arg\max_{\delta \in [-120,\,120]\ \min} \operatorname{corr}\left(p_{\mathrm{pre}},\, \operatorname{shift}_{\delta}(p_{\mathrm{post}})\right). \tag{10}$$

4.3. Labelling rules

Nineteen rules turn these signatures into votes (Table 3). Each rule emits one label or abstains. The rules fall into nine families: simultaneity, holds, restoration, city, single zone, calendar, clock, conservation and straight-line fills. Nine of the rules are *defining*, i.e. their condition is the type's definition. A rule may vote for the grid-event family as a whole (G) rather than a single type. Rule numbers follow the released code: two rules planned on external utility and press records (R17, R18) were not implemented, so no rule uses external records and the numbering skips from R16 to R19.

4.4. Why the standard label model fails here

The Dawid–Skene model [7] treats each rule r as an annotator with a confusion matrix $\theta_r(y, l)$, the probability that r outputs l on an item of type y when it fires, and computes, for item i,

$$q_i(y) \propto \pi_y \prod_{r:\, \lambda_r(i) \neq \varnothing} \theta_r\big(y, \lambda_r(i)\big). \tag{11}$$

When abstentions are ignored, as in Eq. (11), a rule that can emit only one label l has $\theta_r(y, l) = 1$ for every y: its vote multiplies every class by the same factor, and the posterior equals the prior. All nineteen rules are of

Table 3: Labelling rules. Item: the item kind the rule reads. Coverage: share of the real-data items of that kind on which the rule fires. Precision: the label model's learned precision (mean posterior mass on the voted type over the items the rule fires on). R7 never fires on the real data. †Defining rule. r_E, π_E, τ_E: Eq. (8).

Rule	Family	Item	Vote	Condition	Coverage	Precision
R1	Simultaneity	Dead	A1	$S_E \geq 2$, $\eta_E \geq 0.95$, $\lvert r_E - 1\rvert \leq 0.2$, $\tau_E \leq 10$ min	0.008	1.00
R2†	Hold	Fill	A2	Held in ≥ 2 zones	0.077	0.99
R3	Hold	Dead	A1	A hold ends where the dead run starts	0.024	0.67
R4	Hold	Dead	A1	A co-extensive hold in another zone	0.024	0.33
R5	Restoration	Dead	G	Overshoot $\pi_E \geq 1.15$	0.167	0.96
R6	Restoration	Dead	G	Noise floor, $\eta_E < 0.9$	0.667	1.00
R7	Simultaneity	Dead	G1	$S_E \geq 2$, no hold or fill evidence, and R5, R6 or slow restoration	0	–
R8†	City	Dead	G2	$\geq 90\%$ of one city's zones dead, other cities alive	0.008	0.99
R9	Single zone	Dead	G3	One zone	0.968	1.00
R10†	Calendar	Dead	G4	One zone, starts 06:00–11:59, lasts ≤ 6 h	0.548	0.96
R11	Single zone	Dead	G3	One zone, lasts > 24 h	0.032	0.98
R12†	Hold	Fill	A3	Held in one zone, neighbours varying	0.284	1.00
R13†	Clock	Clock	A4	Duplicated stamps on the date	0.242	0.93
R14†	Calendar	Clock	A4	Missing stamps on a clock-change date	0.182	0.70
R15	Clock	Clock	A4	$\lvert\delta^*\rvert \geq 50$ min, correlation ≥ 0.9 at δ^* and ≥ 0.05 above that at 0	0.182	0.89
R16†	Conservation	Transfer	G5	$\kappa \leq 0.3$	0.077	0.68
R19†	Straight line	Fill	A2	Straight-line fill in ≥ 2 zones	0.310	1.00
R20†	Straight line	Fill	A3	Straight-line fill in one zone, neighbours varying	0.310	1.00
R21	Straight line	Dead	A1	A straight-line fill ends where the dead run starts, or is co-extensive	0.040	0.19

this kind, so Eq. (11) cannot use them. Modelling abstention restores the information, but a free confusion matrix per rule is then unstable: most items receive only one to three votes, and different restarts of the estimation reach different labels (Section 6).

4.5. A constrained label model with abstention

Three constraints make the model well-posed and stable. First, each item kind and level has a feasible set $\mathcal{Y}(i)$; for example, a dead item spanning two cities can only be A1 or G1, and a single-zone fill only A3 or N. Second, a rule only counts on the items it can observe (its applicability set $\mathcal{A}(i)$), and its abstention there is informative. Third, each rule output l has a set $S(r,l)$ of types it speaks for. The posterior is

$$q_i(y) \propto \mathbb{1}[y \in \mathcal{Y}(i)]\, \pi_{k(i)}(y) \prod_{r \in \mathcal{A}(i)} \theta_r\big(y, o_{ir}\big), \qquad o_{ir} \in \{\varnothing\} \cup \mathrm{out}(r), \tag{12}$$

with a class prior π_k per item kind, and each output has two parameters, in the spirit of the one-coin model [8]:

$$\theta_r(y,l) = \begin{cases} a_{rl}, & y \in S(r,l), \\ b_{rl}, & y \notin S(r,l), \end{cases} \qquad \theta_r(y,\varnothing) = 1 - \sum_{l} \theta_r(y,l). \tag{13}$$

The expectation–maximisation updates use Beta pseudo-counts:

$$a_{rl} = \frac{\alpha_1 + \sum_i m_i \mathbb{1}[o_{ir} = l]}{\alpha_1 + \alpha_0 + \sum_i m_i}, \qquad b_{rl} = \frac{\beta_1 + \sum_i n_i \mathbb{1}[o_{ir} = l]}{\beta_1 + \beta_0 + \sum_i n_i}, \\ \pi_k(y) \propto \gamma + \sum_{i:\, k(i)=k} q_i(y), \tag{14}$$

where the sums run over the items to which r applies, and m_i and n_i are the posterior masses on $S(r,l)$ and on its complement. We set $(\alpha_1, \alpha_0) = (1,1)$, or $(9,1)$ for defining rules, $(\beta_1, \beta_0) = (1,9)$ and $\gamma = 1$, and keep the best of 20 restarts. The defining-rule prior matters: without it, 39% of the real dead items change label (the sensitivity to every prior and rule family is reported in the supplementary material). With these constraints all 20 restarts reach the same labels, and the model recovers its own parameters from votes simulated from it (Section 6).

4.6. Multi-scale hierarchical residual detector

The detector predicts each zone's relative level $\rho_{z,t}$ (clipped to $[0, 3]$) from information that excludes the zone's own current value, and scores the residual

$$e_{z,t} = \rho_{z,t} - \hat{\rho}_{z,t}, \qquad \hat{\rho}_{z,t} = f\big(\rho_{z,<t},\, \{\rho_{z',t}\}_{z' \in \mathcal{N}(z)},\, \mathbf{c}_t\big), \tag{15}$$

where $\mathbf{c}_t$ holds calendar features and the zone's values one and seven days earlier. The predictor f combines a causal temporal convolutional network [9] over the zone's past (six residual blocks of two causal convolutions with kernel size 3 and dilations 1 to 32, a receptive field of 253 steps, about 42 h) with two-head graph attention [10] over the current values of its neighbours $\mathcal{N}(z)$: the zones of the same city and the four zones of other cities whose learned embeddings are most similar, as in GDN [11]. It is trained with a Huber loss [12] on rule-clean steps only, with two-fold temporal cross-fitting so that every prediction is made out of sample. The learned graph is evaluated as an ablation (Section 6).

The residual is split into frequency bands by a maximal overlap discrete wavelet transform (MODWT) with LA(8) filters [13],

$$W_{j,t} = \sum_{l=0}^{L-1} \tilde{h}_l \, V_{j-1,\, t-2^{j-1}l}, \qquad V_{j,t} = \sum_{l=0}^{L-1} \tilde{g}_l \, V_{j-1,\, t-2^{j-1}l}, \qquad V_0 = e, \tag{16}$$

for $j = 1, \ldots, 4$, where $\tilde{g}$ and $\tilde{h}$ are the LA(8) scaling and wavelet filters rescaled for the MODWT and $L = 8$. Each band is shifted by its group delay so that the bands align in time, and the detector uses $b \in \{e, W_1, \ldots, W_4, V_4\}$. In each band, the residual is decomposed along the hierarchy into a region part, a city part and a zone part:

$$g_t = \operatorname*{med}_{z \in R} e_{z,t}, \qquad u_{c,t} = \operatorname*{med}_{z \in c} e_{z,t} - g_t, \qquad v_{z,t} = e_{z,t} - g_t - u_{c(z),t}. \tag{17}$$

A failure shared by the whole system moves g, a city outage moves u_c and a feeder fault moves only v_z. Each part $\epsilon \in \{g, u_c, v_z\}$ is scored by a trailing robust z-score,

$$s_t^{\epsilon} = \frac{|\epsilon_t| - \operatorname{med}_W |\epsilon|}{1.4826 \, \operatorname{MAD}_W |\epsilon| + 10^{-6}}, \tag{18}$$

where W is the previous 28 days restricted to rule-clean steps and the MAD is scaled to be consistent at the normal distribution [14]. Structural zeros ($|\epsilon| \le 10^{-12}$) are excluded from W: a median over an odd number of zones is

attained by one of them, whose part is then exactly zero and would otherwise collapse the scale.

The zone-step score $Z_{z,t} = \max_b \max(s_t^g, s_t^{u_{c(z)}}, s_t^{v_z})$ is smoothed and thresholded:

$$\tilde{Z}_{z,t} = \frac{1}{w} \sum_{k=-\lfloor w/2 \rfloor}^{\lceil w/2 \rceil - 1} Z_{z,t+k}, \qquad \mathrm{flag}_{z,t} = \mathbb{1}\big[\tilde{Z}_{z,t} > \gamma_q\big], \tag{19}$$

with γ_q the q-quantile of $\tilde{Z}$ on zone-steps more than 53 steps (about 9 h, half the support of the widest band) from any rule state in the zone and from any multi-zone rule state in the system. Flag runs of at least three steps, merged across gaps of up to $g_{\max}$ steps, are the detector's events; those that overlap no rule item are *detector-only candidates*. The setting $(w, q, g_{\max}) = (24, 99.9\%, 12)$, i.e. 4 h smoothing, was chosen on development data like the settings of every compared detector (Section 5). The phase alignment and the centred smoothing use future steps, so the detector is retrospective: an offline catalogue builder, not a real-time alarm.

4.7. End model and fusion

The end model learns the label-model posteriors from features that every item has: structure (duration, numbers of zones and cities), shares of each per-step state, the dynamics of ρ before, during and after the item, calendar, adjacency to fills, episode context and the detector's level and band scores. Rule items carry their posterior q_i as a soft target, normal windows carry the target N, and detector-only candidates are unlabelled. A LightGBM classifier [15] minimises the soft-label cross-entropy

$$\mathcal{L} = -\frac{1}{n} \sum_i \sum_y q_i(y) \log f(y \mid \boldsymbol{\phi}_i), \tag{20}$$

and predicts out of fold, with folds grouped by episode. The fusion is a weighted product of the two experts [16], restricted to the feasible set:

$$p_i(y) \propto \mathbb{1}[y \in \mathcal{Y}(i)]\; q_i(y)^{\beta}\, f(y \mid \boldsymbol{\phi}_i)^{1-\beta}. \tag{21}$$

Each expert carries evidence the other lacks (rule votes; detector and context features), so either can change the winning type. The weight β is tuned on the training period of the injected data (Section 5); the real southern catalogue uses $\beta = 0.6$, selected at 3% contamination. Marrakech lies outside the

injection benchmark and keeps the untuned default $\beta = 0.5$. This is harmless: the two experts agree on 53 of its 55 items, and labels are identical for all $\beta \geq 0.4$; lower values move at most two items from G3 to G4. The end model abstains ($p = q$) on items whose label-model class has fewer than three items of training mass, and detector-only candidates use a uniform q, so their type follows f. Clock and transfer items keep their label-model posterior.

4.8. Type-aware cleaning

Cleaning changes only flagged zone-steps (dead, held or straight-line states inside a qualifying run). Four strategies are compared: S0 keeps the raw data; S1 interpolates every flagged run linearly, outages included; S2, the type-aware strategy, uses the catalogue; S3 is an oracle that uses the true types and exists only for injected data. In S2, artefact steps (A1–A3) are imputed from the expected value, rescaled so that the imputation joins the observed series at both edges:

$$\tilde{x}_{z,t} = \hat{x}_{z,t}\big[(1 - \lambda_t)\,\bar{r}_{\mathrm{L}} + \lambda_t\,\bar{r}_{\mathrm{R}}\big], \tag{22}$$

where $\bar{r}_{\mathrm{L}}$ and $\bar{r}_{\mathrm{R}}$ are the median ratios $x/\hat{x}$ of up to three valid steps within 6 h before and after the gap, and λ_t rises linearly from 0 to 1 across it. Grid-event steps (G1–G5) are kept as recorded, marked by an event indicator that the forecaster reads, and excluded from its training loss. Items labelled N, clock and transfer items and detector-only candidates are left unchanged.

5. Experimental design

Detection and attribution are measured on injected events, and the value of the catalogue downstream.

5.1. Injection benchmark

Events are written into the real southern series inside *clean zone-time*: zone-steps more than 53 steps from any rule state, after a 35-day burn-in and away from clock-change days. Table 4 lists the nine types; two untyped ones probe the detector where no rule fires. Events are placed in random type order, without overlap, until they cover 1, 3 or 7% of the clean zone-steps. The benchmark favours the rules: the injected artefacts copy the episodes from which the rules were written, and the ramp and overshoot of injected grid events are hypotheses that the real data do not confirm (Sections 6 and 7).

Table 4: Injected event types. Durations are log-uniform in the range, except G4 (uniform). $\hat{x}$: expected value (Eq. (1)); $U(a, b)$: uniform draw. Artefact signatures are copied from real episodes; grid-event signatures follow the hypothesised physics of Section 3.4. Each event blocks its zones for ±12 h, and a multi-city event blocks the whole system for ±3 h. Clock errors (A4) and load transfers (G5) are not injected: their rules are calendar-gated or act on whole days.

Type	Zones	Duration	Injected signature
A1	all zones of 2–3 cities	0.5–24 h	exact zeros
A2	2 or more zones of one city (50%), one city (25%) or 2–3 cities (25%)	1–48 h	each zone held at its last value or interpolated by a straight line (50/50); a whole-city fill is followed by an A1 zero fill with probability 0.5
A3	one zone	1–6 h	hold or straight-line fill
G1	all zones of 2 cities	1–18 h	noise floor $\hat{x} \cdot U(0, 0.03)$ with 0–30% exact zeros; linear ramp of 20–90 min; overshoot $\omega \sim U(1.1, 1.5)$ decaying over 1–2 h
G2	all zones of one city	20 min–6 h	as G1
G3	one zone, any hour	1–48 h	as G1
G4	one zone, start 06:00–11:59, 70% at weekends	1–6 h	noise floor as G1; ramp of 10–30 min; no overshoot
U_SHOCK	all zones of 2–3 cities	0.5–4 h	scaled by $U(0.5, 0.8)$; no rule fires
U_PARTIAL	one zone	1–8 h	scaled by $U(0.10, 0.45)$, above the dead threshold

The predictors are not retrained: cross-fitted on the real series, they predict every injected step from the other half of the real data. The rule engine, label model and end model are re-run on each injected dataset as on the real data. Every design choice was made on five development datasets: 3% with seeds 100–102, 1% with seed 103 and 7% with seed 104. The 30 test datasets (three rates × seeds 0–9; 8,526 events) were used for no choice. Results refer to the test period, July 2023 to May 2024; the earlier period serves only to tune β.

5.2. Methods compared

Detection compares the rule engine alone, the main detector alone and the full method (rule items plus detector-only candidates) with three classical baselines, STL [17, 18] with a daily period, scored by the robust z-

score of its remainder (Eq. (18)), and Isolation Forest [19] and LOF [20] on five per-step features of ρ, and with two deep multivariate detectors: GDN [11], attention forecasting over a learned graph and the published method closest to our graph component, and USAD [21], two adversarially trained autoencoders. Both use their authors' released settings and are trained like our predictors. Ablations change one component of the main detector: the learned graph (removed), the predictor (LightGBM), the scales (raw band only), the hierarchy (flat zone score) or the event formation (the untuned setting $(w, q, g_{\max}) = (1, 99.5\%, 1)$).

Every detector, baselines included, receives its own event formation: the best of the same 48 settings of $(w, q, g_{\max})$ by mean event F1 on the development datasets. Ablations inherit the main detector's setting; the no-graph ablation is also reported at its own. The main detector was chosen among four predictors by a criterion fixed beforehand: the development F1 of the rules plus its candidates.

Attribution compares the rules' majority vote, the label model, the end model and the fusion on the typed injected events that the full method detects.

5.3. Metrics

Detection is counted per event, without the point adjustment that inflates scores [22]. With E the injected and $\hat{E}$ the predicted events,

$$P = \frac{|\hat{E}^*|}{|\hat{E}|}, \qquad R = \frac{|E^*|}{|E|}, \qquad F_1 = \frac{2PR}{P+R}, \tag{23}$$
$$\hat{E}^* = \{\hat{e} \in \hat{E} : \hat{e} \cap E \neq \emptyset\}, \qquad E^* = \{e \in E : e \cap \hat{E} \neq \emptyset\},$$

where $\hat{e} \cap E \neq \emptyset$ means that $\hat{e}$ overlaps an injected span in time, its restoration tail included; the zone-strict variant also requires a shared zone. Predictions that touch neither clean zone-time nor an injected span are not scored. A second view also ignores each method's own alarms on the uninjected series, as rule states already are. Step-wise AUC-PR and recall per type are also reported.

Attribution is scored by macro-F1 over the seven typed classes and by the expected calibration error [23] of the arg-max probability over ten equal-width bins B_b:

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{n} \left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|. \tag{24}$$

Forecasts $\hat{y}_{z,t}$ are scored on a set S of zone-steps by the mean absolute scaled error [24],

$$\mathrm{MASE} = \frac{1}{|S|} \sum_{(z,t)\in S} a_{z,t}, \qquad a_{z,t} = \frac{|x_{z,t} - \hat{y}_{z,t}|}{s_z}, \tag{25}$$

where s_z is the mean absolute one-day change of zone z on normal training steps. Two strategies are compared by a Diebold–Mariano test [25] on the loss summed over zones, $L_t = \sum_z a_{z,t}$:

$$d_t = L_t^{(1)} - L_t^{(2)}, \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \qquad \hat{\Omega} = \hat{\gamma}_0 + 2\sum_{k=1}^{h} \Big(1 - \frac{k}{h+1}\Big)\hat{\gamma}_k, \tag{26}$$

where $\hat{\gamma}_k$ is the lag-k autocovariance of d_t, with the Newey–West bandwidth [26] equal to the horizon h because direct forecasts overlap. Outage counts are compared by the ratio of counted to true outage zone-hours and its log distance:

$$r = H/H^{\mathrm{true}}, \qquad \delta = |\log r|. \tag{27}$$

5.4. Downstream test

On the southern system, the cleaning strategies S0–S3 (Section 4.8) feed two forecasters, 1 h and 1 day ahead: a global LightGBM direct forecaster that also reads the event indicator, trained up to 31 December 2023, and a model-free reference (persistence at 1 h, seasonal naive at 1 day). Forecasts are scored from 1 January to 24 May 2024 against the recorded values, on zone-steps that every strategy treats as normal, and separately on those within two days after an anomalous state. Forecasts and outage counts are evaluated on the real data and on all 30 injected test datasets. Naive counting takes every flagged dead state and typed counting only those inside grid-event items; the truth is the injected G1–G4 spans.

5.5. Transfer test

Tetouan tests the method on a new system with nothing re-tuned. The rules, the label-model parameters (applied by one E-step), the architectures, the training settings and the event formation are frozen; the expected value, the score scales, the predictor weights and γ_q (at the frozen percentile) are re-estimated from Tetouan's unlabelled series. With one city, G1 is not injected and multi-city types become city-wide. Thirty fresh injected datasets (three rates $\times$ 10 seeds) are scored over 2017.

5.6. Statistics and reproducibility

Results are mean ± standard deviation over the 10 seeds of each rate. Methods are compared by exact paired two-sided Wilcoxon signed-rank tests [27]: the smallest attainable p is 0.002 over 10 seeds and 1.9×10^{-9} over 30 datasets. The code uses Python, PyTorch [28], LightGBM 4.6.0 [15] and scikit-learn [29]; every injected dataset is regenerated from its rate and seed, and versions, settings and run times are given in the supplementary material.

6. Results

[To be written.]

7. Discussion

[To be written.]

8. Conclusions

[To be written.]

Data and code availability

The UCI datasets analysed in this study are publicly available [1, 4]. The Moroccan data are the file `Data Morocco.xlsx` of the UCI record (22,434,854 bytes; SHA-256, in groups of eight hexadecimal digits: `507134d6 7ecbbddb f17640e1 937da132 d0db8000 fbfd32a8 8dda9a4c bf53c6fc`), byte-identical to the file served by the repository on 14 November 2026. The code, the typed event catalogue and all results will be made publicly available. [Repository and archive identifiers to be added.]

CRediT authorship contribution statement

[To be added by the authors.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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